Problem Chosen	2020 MCM/ICM Summary Sheet	Team Control Number
F		2016882

Embrace Human Rights and Culture in New Home

Climate change is expected to affect habitability in many parts of the world in significant and unprecedented ways in the coming decades. The rise in sea level means the great challenge that nature gives us: For environmentally displaced persons (EDPs), where will they go, can their human rights be guaranteed, and can their culture be protected and inherited? We have established a series of models and its corresponding policies to try to solve the above problems.

Firstly, we divided the future climate changes into three scenarios: optimistic, retain, and pessimistic. We used different functions to represent the relationship between sea level anomalies and time in these scenarios. Next, we constructed the **Island Elevation-Resource Model** based on the natural and socio-economic factors of the island. In this way, we analyze the number of people at risk of becoming EDP over time. We propose "**specificity**" as an important factor for the risk of cultural loss. We believe that the risk of cultural loss with a small specificity is also small, and resettlement issues should be given priority when relocating; The risk of cultural loss in highly specific cultures is closely related to where they migrate. We take the Maldives and the Marshall Islands as examples for analysis.

Secondly, we use the **TOPSIS Method** to construct the **Attractiveness-Obligation- Difference (AOD) Comprehensive Evaluation Model** with the three indicators of regional attraction, international obligations, and cultural differences. We use a regional gravity model that is innovative to characterize the degree of fit between the relocating country and the receiving country. For international obligation indicators, we use the cumulative carbon dioxide emissions to measure. For cultural differences, we propose the cultural vector model. Our model can select the most suitable recipient countries for a particular relocated country to fully meet the human rights and cultural protection needs of the people in the relocated country. We also provided the UN with a guide to action to address long-term or timely relocation needs, which includes an early warning application mechanism, a recipient country selection process based on the AOD model, and ongoing safeguards. We take Maldives as an example to provide relocation guidelines for this country.

Thirdly, although the AOD model has taken cultural differences into consideration, even small differences will inevitably lead to cultural exchanges, conflicts and assimilation, which will have a potential impact on the culture of the relocated country. To this end, a **cultural blending model** was established. We used some points to represent individuals, and the encounter between points to represent cultural exchanges, and also specified the probability and degree of cultural assimilation. We performed a simulation analysis using Maldives' migration to Hubli, India as an example, and obtained the conclusion that the Maldivian culture will tend to die out around 2300.

Fourthly, we use the **cultural integration model** to conduct simulation analysis on cities, living patterns, and assimilation probabilities of different populations, resulting in the following three important recommendations: Under the same conditions, refugees should migrate as close to or less as possible to their population To live in cities; cluster living can better protect refugee culture than scattered living; improve the national cohesion and cultural confidence of refugees.

In addition, we explain the importance of implementing our proposed policies as follow: guarantees both individual and collective rights; combines the capacity for nations and the obligations to climate change; strikes a good balance between protecting the culture and reducing cultural conflict.

At last, we analyze the stability and sensitivity of our model. We also conclude the strengths and weaknesses of it. Further, more time factors such as rising nationalism should be consider to make our model and policies better.

Key words: EDPs, AOD Model, cultural communication simulation, cultural integration model

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1 Introduction

1.1 Background

Researchers have found that in recent years, with global warming, sea levels have continued to rise and flooded coastal areas. Globally, some island nations, such as the Maldives, Tuvalu, Kiribati and the Marshall Islands, are in danger of disappearing completely. The population of these island nations will lose their homes, and these people who lost their homes are called environmentally displaced persons (EDPs). The immigration of these island nations is imperative, and a series of problems will follow. What countries should the population of these island countries migrate to in the future? What countries will accept these immigrants？ Whether other countries have obligations to accept these immigrants or not? What will happen to human rights and cultural protection?

1.2 Restatement and Our Work

As a result of a recent UN ruling that opened the door to the theoretical recognition of EDPs as refugees, the International Climate Migration Foundation hires us to analyze this multifaceted issue of when, why, and how the UN should step into a role of addressing the increasing challenge of EDPs.

In this paper, we are mainly going to finish the work below:

- **Part 1:** Analyze the scope of the issue in terms of both the number of people at risk and the risk of loss of culture.
- **Part 2:** Provide the United Nations with human rights and cultural protection policies for EDP issues.
- **Part 3:** Build Cultural integration model to describe the potential impact of the policies we provide.
- **Part 4:** Use the potential impact models to improve policies.
- **Part 5:** Based on the analysis, explain the importance of implementing the proposed policy.

2 Assumptions

In order to draw some reasonable conclusions from our model and make it convenient for us to simulate real-life condition, we make the following basic assumptions, each of which is properly justified.

- Assume that our research object is a smaller island nation. If a large country’s coastal city is inundated due to rising sea levels, the population of the coastal city will not be included in the EDPs, because the city's population can be migrated in land to the country and can be resolved without the intervention of the United Nations.
- Assume that for the same ocean, the speed of sea level rise is the same in all of its sea areas. Since there is a seasonal dynamic balance between sea levels in the ocean, this assumption has little effect on the accuracy of the model and can make the model more simplified.
- Assume that the number of people who need to be relocated in a certain country in the future is the national population in 2019 and the population is evenly distributed across the country. Since the countries that facing the risk of environmental migration are all small countries with small national population, so the assumption has little impact on the accuracy of the model and can also simplify calculations.
- Assume that cultural exchanges and assimilation only occur when people actually contact each other. We ignore the assimilation caused by cultural influences such as film and television since they have far less impact than the former.

3 Notations

Notations	Definitions
ΔH_0	the initial sea level in one area
r	the growth rate of sea level
sl_d	the sea level danger alert line
R_i	the existing resource ratio
G_a	the per capita GDP
h_i	the average altitude of the island nation
L_{lack}	labor shortage in host government
L_{in}	workforce in relocated country
D	the distance between two countries
O_i	international obligation in country i
p_i	CO_2 emissions from country i
ω	the culture vector
CP	cultural protection indicator
$Status_i$	the status vector of person i
v_i	the velocity vector of person i

4 Part One: Analyze the Population at Risk and the Risk of Loss of Culture

4.1 Environmentally Displaced Persons

In recent years, due to the long-term turmoil in the Middle East, a large number of refugees have been caused. These people are called war refugees. After these refugees fled to other countries, their human rights were not protected and they were in a vulnerable position.

Researchers have discovered that several island nations, such as the Maldives, Tuvalu, Kiribati, and The Marshall Islands, are at risk of disappearing entirely due to rising sea levels and there are lots of environmentally displaced persons (EDPs) need to relocate. These people have been forced to move to other countries due to meteorological problems caused by rising sea levels. They are not the same as war refugees and their human rights and cultural protection need the attention of the United Nations.

4.2 Model of Rising Sea Level

We know that whether it is an unexpected disaster or a long-term environmental change, it may bring about temporary or permanent forced migration. In actual researches, we can't accurately predict unexpected natural disasters in advance, so we only consider the long-termly occurring sea level rising in this article.

From the knowledge of meteorology, we know that due to the large amount of carbon dioxide produced by human beings in daily life and production activities, global temperatures have risen and Antarctic and Arctic glaciers have melted, leading to global sea level rise. This has led to a large number of people of island nations have to find another place to live. The whole process diagram is as follows:

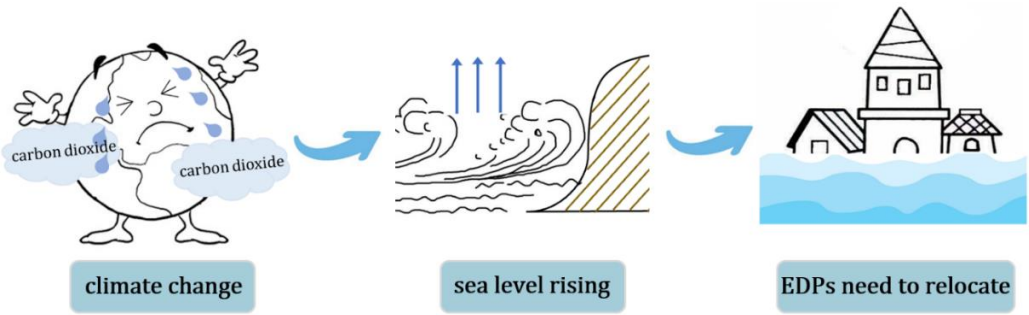


Figure 1. the whole process

With global warming and rising sea levels, we are full of worries about countries that may face disappearance in the future. Estimating in advance where and how many people will be at risk of becoming EDP will help us better help them immigrate. We use the TP / Jason-1 and ERS / ENVISAT AVISO (Archiving Validation and International of Satellites Oceanographic Data) [1] provided by the French National Centre for Space Research to use sea level anomalies (SLA) from 2005 to 2019 data to predict future sea level altitude trends. First, we divide the global sea area into the Pacific Ocean, the Atlantic Ocean, and the Indian Ocean according to traditional definitions and ignore the local sea areas inside the continent, as shown in Figure 2. Selected coverage range is 0-360 ° E, 60 ° S-70 ° N.

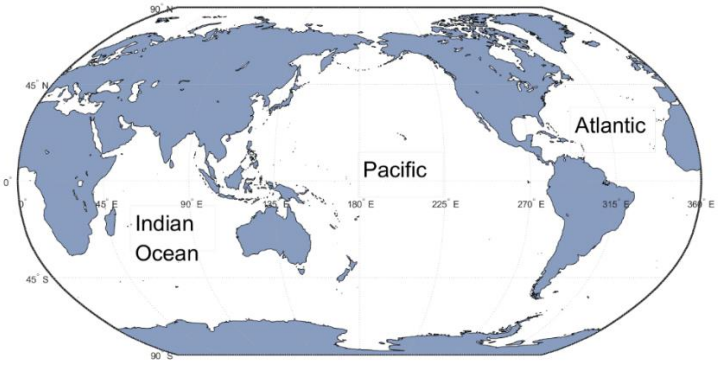


Figure 2. Division of global seas

For the convenience of calculation, we adjusted the original ocean boundary to a regular pattern arranged horizontally and vertically in latitude and longitude. We first average the sea level anomalies within the delimited range, and use this value as the sea level anomalies value of this ocean during the year. To address the uncertainties of analyzing migration over the next 30 years, we consider three potential climate and development scenarios. The scenario-based results should be seen as a plausible range of outcomes rather than precise forecasts. The three scenarios are:

We define ΔH_0 is the initial sea level in this area, r is the sea level growth rate and Δt is the year.

For **"climate-pessimistic"** scenario, which is high greenhouse gas emissions combined with unequal development pathways. We liken it to Malthusian growth model， which is widely regarded in the field of population ecology as the first principle of population dynamics[2]. Our climate-pessimistic model has the following form:

$$\Delta H(t) = \Delta H_0 e^{r\Delta t} \tag{1}$$

For "**climate-retain**" scenario, which is similarly high emissions but with improved development pathways. We use simple linear function fitting as in [3], since under this assumption the rate of development to sea level altitude will remain unchanged. Our climate-development model has the following form:

$$\Delta H(t) = r * \Delta t + \Delta H_0 \tag{2}$$

For "**climate-optimistic**" scenario, which is lower global emissions combined with equal development. Under this scenario, with the advancement of technology and the enhancement of human

environmental awareness, climate change will gradually ease. Through the above analysis, we use the horizontal line to fit sea level data. Our climate-optimistic model has the following form:

$$\Delta H(t) = \Delta H_0 \tag{3}$$

The result of rising sea level model is shown in Figure 3.

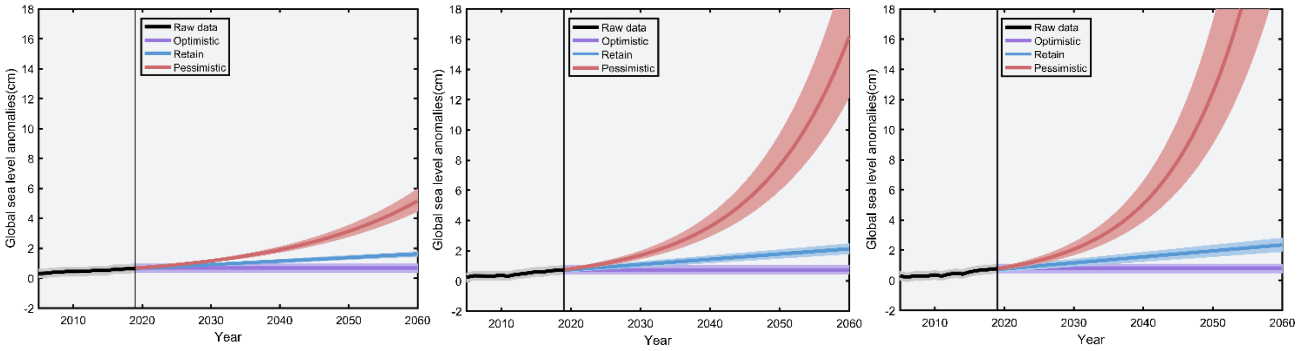


Figure 3. Global sea level anomalies
(from left to right are The Pacific Ocean, Indian Ocean and Atlantic)

As we can see, the sea levels in the Pacific, Indian Ocean, and Atlantic Ocean continue to increase over time in the three modes. Among them, the sea level increase of the Atlantic Ocean is the fastest among the three oceans mentioned above. In addition, we also found that the sea level growth rate in the climate-pessimism model was much faster than the other two models. In other words, if countries can strengthen environmental protection, such as reducing carbon dioxide emissions, they can effectively long-term down sea level rise, which deserves our high attention.

4.3 Countries at Risk

We visualize such sea level anomaly data to have a better understanding of its growth, the global sea level anomalies in 2021 and 2024 is predicted under climate-pessimistic scenario. As shown in Figure 3, we find that the rate of sea level anomalies is accelerating, and more and more countries are facing danger. Changes in sea level anomalies—currently averaging three millimeters a year and increasing—will be the most problematic for settlements on very low-lying land (within one meter of sea level), especially areas that experience subsidence, erosion, or tropical cyclones and extreme storm events. Sea level rise can salinize groundwater and surface freshwater bodies. It can also increase salt-water intrusion into freshwater aquifers (particularly in the Middle East and North Africa), a process made worse by other climate impacts (such as reduced water availability). Together, these impacts present substantial problems to agriculture and human consumption.

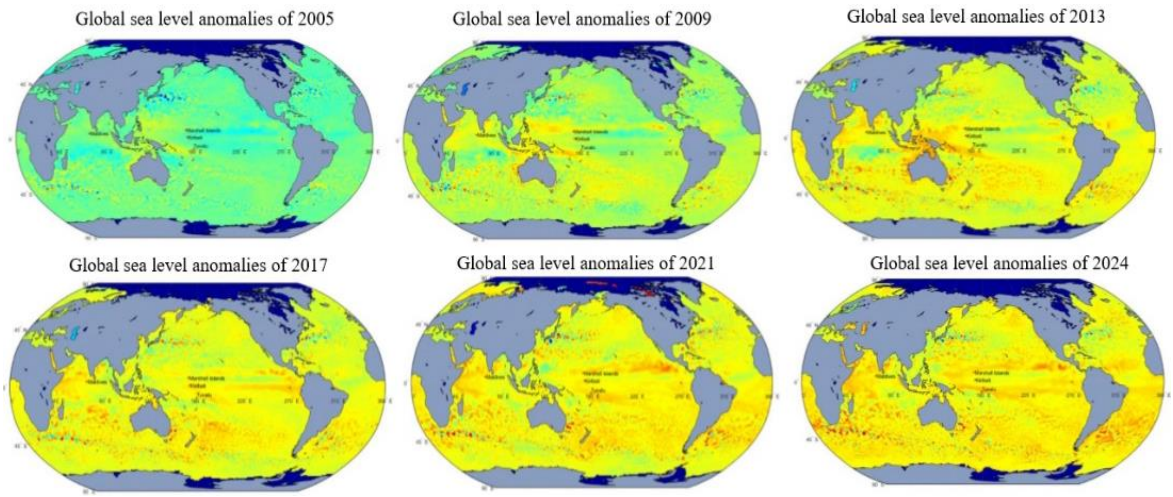


Figure 4. Global sea level anomalies change from 2005 to 2024 under climate-pessimistic scenario

Since the define of Environmentally displaced persons (EDPs) is the people who must relocate as their homeland becomes uninhabitable due to climate change events. According to the geographical environment of each country, some countries may have the possibility of domestic migration. We do not consider this situation here. We only consider national migration due to sea level and its secondary disasters. Let's first look at the low-altitude island nations, as they are the most easily submerged. Our research objects are the Maldives, Tuvalu, Kiribati and the Marshall Islands. We combine the sea level of these countries with the future sea level trend curve, and we get Figure 4.

4.3.1 The Number of People at Risk

4.3.1.1 Island Elevation-Resource Model

We define the "**sea level danger alert line**" sl_d as the minimum sea level height at which a country needs environmental immigration, and use it to link sea level rise to the needs of environmental immigration. We know that with the rise of sea level, the land area of the island nation, natural resources such as fresh water will become less and less. When the island nation is completely submerged, the island's human-friendly resources completely disappear. Here we define the existing resource ratio R_i as the ratio of the existing resources of an island nation to the original total resources of the island nation. When the sea completely submerged the average elevation of the island, we consider that R_i is equal to zero. The reduction rate of R_i is related to the sea level rise value sl, the average altitude of the island nation h_i, and the per capita GDP G_a.

In order to express the quantitative relationship between R_i and h_i, G_a, we represent h_i, G_a as the coefficient of sl. As the sea level rise value sl increases, the decreasing speed of G_a and R_i shows a long-terming trend. Since we can not directly describe the correlation between natural disasters and economic levels, we first set the coefficient of G_a to α times of h_i. Therefore, the existing resource ratio R_i can be calculated as follows:

$$R_i = 1 - \sqrt{sl(1 + 1/h_i)(1 - G_a/\alpha)} \tag{4}$$

Moving the resource ratio R_i of this formula to the right and the sea level rise value sl to the left, we can get:

$$sl_d = \frac{{R_d}^2}{(1 + 1/h_i)(1 - G_a/\alpha)} \tag{5}$$

Here, sl_d represents the sea level danger alert line, R_d represents the existing resource ratio danger alert line. For h_i, we obtained the average altitude of each country through Google Earth. For G_a, we obtained the per capita GDP of each country through World Bank (World Development Report 2018).

Let's first set up that the island nation needs to move when the existing resource ratio R_i reach the danger alert line $R_d = 0.25$, because the amount of resources is far from sufficient to support people's lives. Also set the environment-economic ratio parameter α to 10.

4.3.1.2 Predict Immigrants in Future

In order to test the rationality, validity, and accuracy of the model and preset parameters, we use the case of Maldives to do model testing. Substituting the parameters we set and the obtained data into the above formula, we get that the sea level danger warning line sl_d when the Maldives needs to be relocated is 0.34m. Based on the above optimistic forecast of future sea level rise, we conclude that Maldives will immigrate in the country in approximately 48.5 years. This prediction result is basically consistent with [4], so we think this model has certain reliability.

Below we use this model for flooding time prediction in other island nations. We combined the above forecast model of future sea level rise with the **island elevation-resource model** here to obtain the **sea level danger alert line** and the specific time and year of immigration for each island nation.

Table1. Forecast dates of immigrants

Island nation	Average altitude (meter)	GDP per capita (dollar / per)	Sea level danger alert line (meter)	The year of immigration			The number of EDPs (10,000 people)
				optimistic	retain	pessimistic	
Maldives	1.2	10224	0.34	2068	2049	2040	52
Marshall Islands	2	3621	0.388	2079	2057	2048	6
Kiribati	1.7	1625	0.359	2075	2056	2047	12
Tuvalu	1	3701	0.291	2065	2051	2044	1

Four island nations are predicted to be flooded in three climate scenarios as shown below:

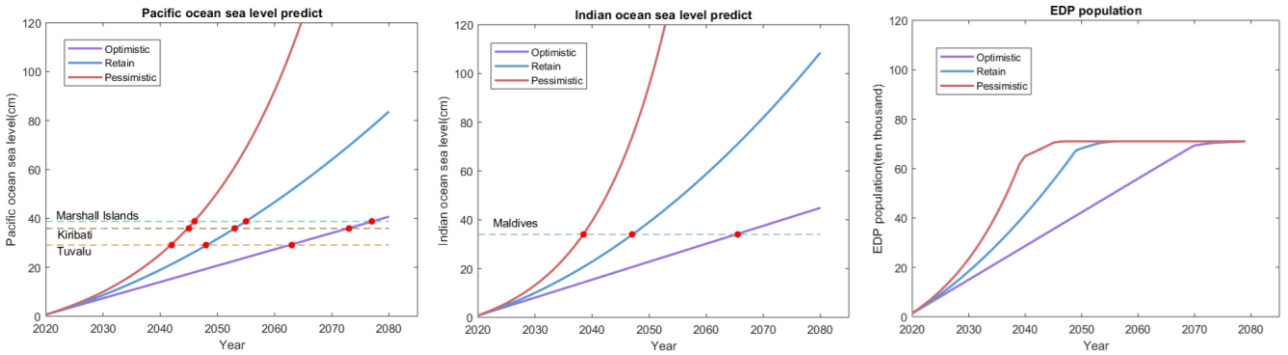


Figure 5. Extinction time and EDP population forecast for four countries

We draw a graph of the total number of EDP changes in the four countries over time, we believe that the island nation's resource ratio R_i reaches the warning line R_d, the island must be relocated by the country; before reaching the R_d, we assume migrant population is a certain ratio of the total population of the island nation。

4.3.2 The Risk of Loss of Culture

As the sea level rises, the entire population of some countries has to be transferred to other countries. This is a great ordeal for the cultural protection of island nations. Thus, how do we view the risk of loss of culture?

We all know that culture is specific, and different countries have their own characteristics. Even the leading nations, their cultures such as language, religion, festivals, art, production and lifestyle may be very different. In this regard, when we study different island nations, we need to pay special attention to the "**specificity**" of the island nation 's culture. If an island nation's cultural characteristics are unique in the world, this country will face huge cultural loss after immigration, such as assimilation of language by local residents, and changes in production and lifestyle due to regional reasons. For example, countries that make a living from fishing will undoubtedly lose their fishing culture if they move to mountains.

Therefore, we can divide the island nations that may be submerged in the future according to the "specificity" of culture, and divide them into two categories， which are countries with low specificity and countries with high specificity.

- Countries with **low specificity**: The risk of cultural loss in such countries is small, and we give priority to population resettlement and human rights.

Cases with low risk of cultural loss: **Maldives**, an archipelago country in the Indian Ocean. Although it is an island nation, it is not the same as other indigenous islands. Food in Maldives is mainly fish, coconut, and cassava. With the development of the economy, food culture is influenced

by South Asian countries, and import food such as rice and flour. Traditional music and dance are deeply influenced by East African culture, such as drumming and music. The cultural specificity of Maldives is not very strong. If the Maldives islands are submerged, the resettlement problem can be solved first and you can choose to relocate to other tropical islands in South Asia. In terms of cultural protection, because the country's cultural specificity is not strong, the risk of cultural loss is small. [5]

- Countries with **high specificity**: The risk of cultural loss in such countries is greater. Our decisions affect the risk of cultural loss. In order to minimize cultural loss, we need to make reasonable decisions to the United Nations. The main consideration is the protection of culture.

Cases of high risk of cultural loss: **The Marshall Islands**, an island nation in the Pacific Ocean. Its main source of income is overseas aid. The people ’s life is mainly based on marine fishing and some traditional handicrafts. If the Marshall Islands moves nationally due to rising sea levels, due to its unique marine culture, the risk of cultural loss is greater. Therefore, cultural protection must be considered first, and it should be placed on an island nation similar to the culture of the Marshall Islands to protect the fishing culture and craft culture of the Marshall people. [6]

5 Part Two: Guidelines for the United Nations

In this task, we are wanted to propose policies to address EDPs in terms of both human rights and cultural preservation.

Based on the results we have learned, it is not difficult to find that in the long run, the population of EDPs will continue to grow, and the island’s migration needs will continue to rise. A small number of countries can complete population migration through independent consultations with other countries, but most countries need the intervention of the United Nations to negotiate with the international community. In the role of the United Nations as a middleman, how to choose the host government for the island nation is a crucial part.

5.1 AOD Comprehensive Evaluation Model

When a country has exact migration needs and cannot solve the problem on its own, the intervention of the United Nations will be necessary. The United Nations should have its own way to initially prepare a list of host governments, and this process involves a comprehensive evaluation model.

A more suitable host government should have a higher economic level and ample living space, and the labor gap and immigrant labor of the host government will become important factors for mutual cooperation between the two countries. Secondly, due to the consideration of the fairness of the international environment, the global climate Countries with greater contributions to warming should also assume corresponding international obligations. In addition, the human rights and cultural protection of environmental refugees remind us to avoid cultural conflicts between receiving and migrating countries as much as possible. It is another key factor in the choice of host governments. Therefore, we give an AOD comprehensive evaluation model, including three aspects: **regional attractiveness**, **international obligations**, **differences of human rights and culture**.

5.1.1 Regional Attractiveness

In fact, there is a push-pull relationship between host governments and relocated countries [7], which means that they all want to maximize their own interests: Nationals of environmental refugee countries hope to live in better areas and meet their work, study and living needs, which means that their human rights can be guaranteed. The host government hopes that this incident will not affect the normal life of its nationals, and will help the country's economic development to a certain extent. For example, those countries with large labor gaps want to absorb more young labor. This interactive push-

pull relationship makes us intuitively associate the gravity model. Gravity model originates from the Gravity Theory, indicating the attraction between two related subjects. The basic formula is:

$$F = G\frac{M_1 M_2}{D^2} \tag{6}$$

G is the constant coefficient, $M_1 M_2$ is the product of "attractiveness" of two country, representing the mass product of two country, r is the distance between two countries. This model is widely applied to measure the economic and financial attraction. I. S. Lowry also developed a regression model (Lowery, 1996) which shows that job opportunity attracts people to migrate from one place to another.

◆ host government's attractiveness M_1

Now we are to measure the weight for the attractiveness inspired by those models to help UN make better decision on migration for environmental refugee. We revise our model by figuring out a comprehensive weight indicator to represent the attractiveness to both moving in and out of the country. To calculate the compatibility of immigration between the two countries, we apply the metrics developed above, including GDP per capita, labor shortage and national happiness index of migrating countries, workforce of relocated countries, and the ratio of relocated population to the population of host governments. Now add the variables derived from the metrics above to M_1, we get:

$$M_1 = G_{ave} L_{lack} I \tag{7}$$

The specific meaning of each notation is shown below:

- G_{ave}: Per capita GDP

Per capita GDP is an effective tool for people to understand and grasp the macroeconomic operating conditions of a country or region. It is often used as an indicator of economic development in development economics and is one of the most important macroeconomic indicators. A country with a higher level of economic development is more suitable for absorbing the relocated population.

- L_{lack}: Labor shortage

The shortage of labor resources includes both quantity and quality. As many countries continue to age, labor shortages are severe. This means that people in relocated countries have more jobs, which is attractive to them.

- I: The national happiness index

The national happiness index refers to the national identity, sense of belonging, stability, and satisfaction of the country, as well as the yearning and praise of the outside population. A country with a high happiness index is often attractive to the relocated population.

◆ relocated country's attractiveness M_2

Add the variables derived from the metrics above to M_2, we get:

$$M_2 = \frac{L_{in}}{P_{in}/P_{ori}} \tag{8}$$

- L_{in}: Workforce

The relocated country's labor force helps alleviate the labor shortage problem in the immigrating country. Adequate labor resources are attractive to the immigrant country.

- P_{in}/P_{ori}: The ratio of relocated population to the population of host governments.

This indicator measures the relative number of people between the two countries. Since the immigrant country does not want a large number of foreign populations to enter, a suitable ratio will help increase the enthusiasm of the immigrant country to accept the relocated population.

- **the distance between two countries *D***

Distance plays a crucial role in migration decisions. On the one hand, distance is a reflection of cultural similarity, and close countries are more likely to have similar customs, practices, art, and values. Close migration can protect the culture of environmental refugees to the greatest extent. On the other hand, short-distance migration can reduce the transportation cost of migration and improve the safety and reliability of migration.

In summary, our regional attractiveness model shows the degree of compatibility between environmental refugee countries and immigrant countries. A high degree of cooperation contributes to human rights and cultural protection.

The indicators used in this model are shown below:

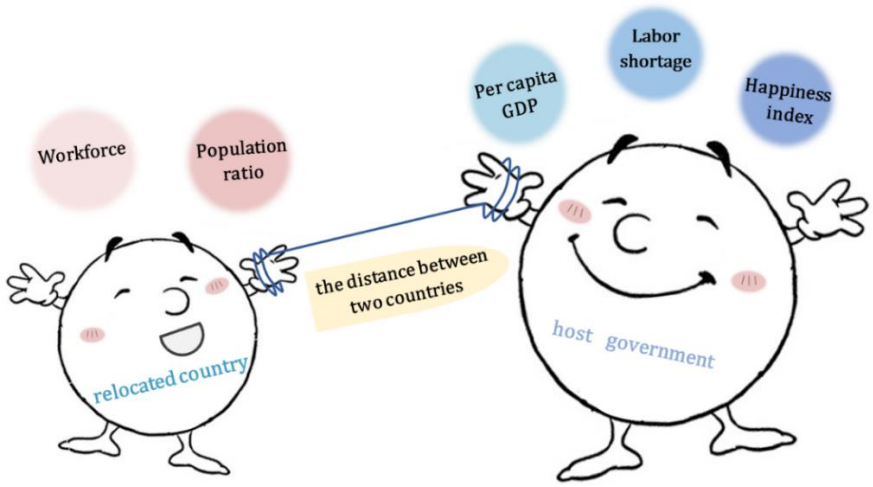


Figure 6. Regional Attractiveness

5.1.2 International Obligation

Given various nations' disproportionate contributions to the green-house gasses both historically and currently that have accelerated climate change linked to the rising seas, we believe that worst offenders have a higher obligation to address these issues.

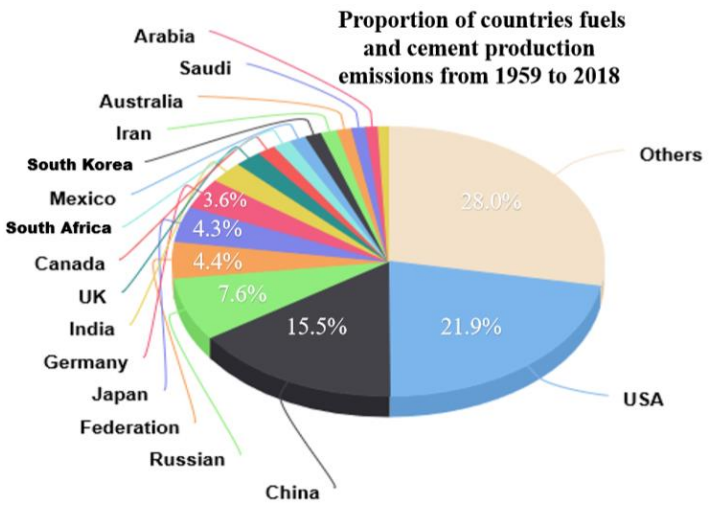


Figure 7. Proportion of countries fuels and cement production emissions from 1959 to 2018

We obtained Fossil fuels and cement production emissions by country from 1959 to 2018 at the Carbon Dioxide Information Analysis Center (CDIAC) [8], and displayed the major carbon dioxide emitting countries in a visual form. We use nearly sixty years of historical data because we believe that the impact of greenhouse gases such as carbon dioxide on global warming is a long historical process. In this historical process, countries with large total carbon consumption need to take more responsibility for global warming. In order to link carbon dioxide emissions with international obligations to undertake environmental migration, we propose an indicator of international obligations for environmental migration in country i as O_i:

$$O_i = \sum_{j=1959}^{2018} p_i^j \tag{9}$$

Where p_i^j is CO_2 emissions from country i in year j. Accumulating the global share of carbon dioxide emissions from country i from 1959 to 2018, we can get the international obligation indicator of environmental migration O_i.

5.1.3 Difference of Human Rights and Culture

The migration of environmental refugees raises two key issues: human rights and cultural protection. The unique culture of the migrating country is an integral part of world culture. Therefore, the choice of a host government with less cultural differences can reduce cultural conflicts, thus can preserve under the aboriginal nationalism. Culture is a difficult concept to quantify, but language [9] and religion [10] dominate the culture. Human rights represent that refugees can resettle in their new homes and fully participate in life. Cultural differences also have a huge impact on them. In addition, differences in the lifestyles of the two peoples will also affect the adaptability of refugee lives. It can be reflected to a certain extent by the difference in economic level [11].

In order to better represent the impact of differences in human rights and cultural protection, we introduce the following **cultural vector model**.

First, we can classify languages and religions into M language systems $L = \{l_1, l_2, \ldots, l_M\}$ and N religion $R = \{r_1, r_2, \ldots, r_N\}$, and we use Distance matrix $A = (a_{ij})$ and $B = (b_{ij})$ to represent the differences between any two languages and religions. When multiple religions or languages coexist, the distance between them can be expressed as a weighted average. In view of the difference in economic levels, we can use the difference d_{GDP} of the GDP per capita of the two countries to measure. From this we can get a cultural distance vector:

$$\omega = (\omega_1, \omega_2, d_{GDP}) \ \ with \ \ \omega_1 \in A, \omega_2 \in B$$

It is easy to find that if the total level of cultural differences between the two countries is fixed, the difference in one of the three causes a greater cultural conflict than the uniform distribution of the differences. Thus, we propose human rights and cultural protection indicators CP:

$$CP = \|\omega\| \cdot \big(1 - cos(\theta + \varphi)\big) \tag{10}$$

Where $\|\omega\|$ is the modulus of the vector ω, which represents the absolute difference; $1 - cos(\theta + \varphi)$ represents the effect of extreme conditions when the absolute difference is fixed; θ represents the vector ω and the ideal direction The angle of $(1,1,1)$ can be determined according to the maximum possible value of θ we can get $\varphi = arccos\frac{\sqrt{3}}{3}$. From the above formula, we can know that the larger the index CP is, the higher the two cultural values are. The larger the difference, the less suitable it is for migration.

5.1.4 TOPSIS Method

From the above analysis, it can be seen that the determination of the host government is mainly based on the three-level indicators of regional attractiveness, international obligations, human rights and cultural differences, which are comprehensively evaluated, including labor gap, migration distance, and population density Secondary indicators. The overall level of the comprehensive evaluation can be seen in Figure 8.

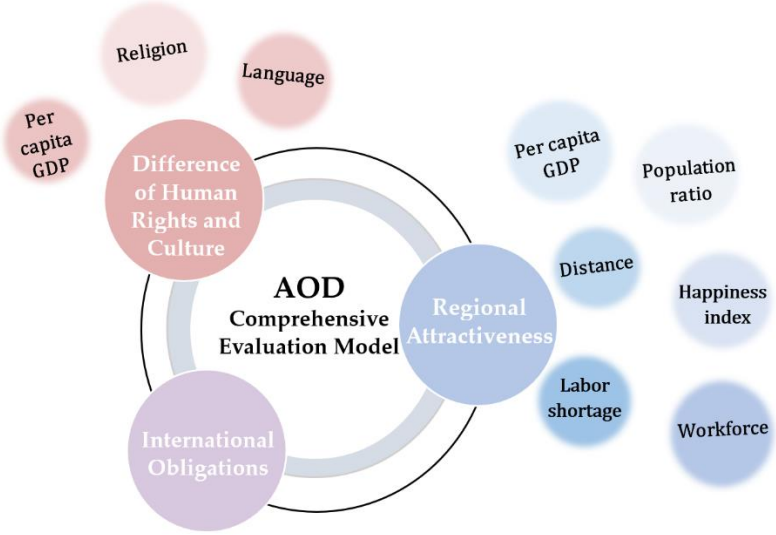


Figure 8. AOD Comprehensive Evaluation Model

The determination of the host government is a relatively important decision, and the weight of the indicator has a decisive influence on the final result. Therefore, in order to avoid wrong decisions caused by unprofessional weights, we choose to use the TOPSIS method [12] that makes full use of the original data. First, we have to preprocess the data.

- **Data Preprocessing**

Same trending: In the first-level indicator, geographical attractiveness and international obligations are positive indicators, and the difference between human rights and culture is a negative indicator. The latter is linearly trended:

$$x_i = \frac{x_{max} - \hat{x}_i}{x_{max} - x_{min}} \tag{11}$$

Standardization: n evaluation objects, 3 evaluation indicators (which are same trending) can form a $3 \times n$ positive data matrix $X = (x_{ij})$, and the following standardization can be obtained the normalized data matrix $Y = (y_{ij})$.

$$y_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{n} x_{ij}^2}} \tag{12}$$

- **Specificity Coefficient**

The degree of specificity of the culture of the migrating country will affect the priority of resettlement and cultural protection during migration, that is, as mentioned at the end of question 1, countries with low cultural specificity will give priority to resettlement during migration, so we introduce a specificity coefficient And update the normalization matrix to $Z = (z_{ij})$, as follows:

$$Z = \begin{pmatrix} y_{11} & y_{12} & y_{13} \\ y_{21} & y_{22} & y_{23} \\ \vdots & \vdots & \vdots \\ y_{n1} & y_{n2} & y_{n3} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & s \end{pmatrix}$$

- **Determine the Ideal Point and Distance**

Based on the data matrix Z, we can determine the positive and negative ideal points are V^+ and V^-. And we can calculate the distance from each country to the positive and negative ideal points:

$$D_i^+ = \sqrt{\sum_{j=1}^{3} \left(V_j^+ - z_{ij}\right)^2}, D_i^- = \sqrt{\sum_{j=1}^{3} \left(V_j^- - z_{ij}\right)^2} \tag{13}$$

- **Comprehensive Evaluation**

Using the distance to the positive and negative ideal points, the relative ideality S_i of each selection scheme can be calculated and sorted according to its value.

$$S_i = \frac{D_i^-}{D_i^+ + D_i^-} \tag{14}$$

5.1.5 Example

Let's take Maldives for example. When this country must make a decision on environmental migration, which country will be the best choice? For reference, we selected Sri Lanka, Australia, India, Singapore, Thailand, Saudi Arabia, and Ethiopia as alternative migration programs based on factors such as distance, economic development level, ageing degree, cultural differences, etc. to simulate the immigration situation that how should the United Nations guide the choice of relocating countries. We obtained the data mentioned in the AOD model of these candidate countries from the World Bank [13], such as GDP per capita, ageing, total number of people, happiness index, etc. Then normalize the same data to avoid the impact of the dimension on the model, and finally put these data into the AOD model. We conclude from our model that after comprehensive consideration of human rights protection, cultural protection and international obligations, the comprehensive ranking of these countries is shown in Table 2:

Table 2. AOD model for Maldives

Country	Selective ranking	Regional Attractiveness Score	Fossil fuels emissions by country from 1959 to 2018(million tons)	Religion	Language
India	1	97.11	13189.98	Hindu, Islam, Sikh	Hindi, English
Sri Lanka	2	37.13	121.13	Buddhism, Hinduism, Christianity, Islam	Sinhalese, Tamil
Saudi Arabia	3	56.38	3909.90	Islam-Sunni	Arabic
Singapore	4	22.24	531.06	Buddhism, Taoism, Islam, Christianity, Hinduism	English, Malay, Mandarin, Tamil
Australia	5	17.62	4339.71	Protestant, Catholic	English
Thailand	6	26.81	352.21	Theravada Buddhism	Thai
Ethiopia	7	9.94	60.87	Islam, Orthodox	Amharic

In our model, India is the most suitable choice. It has higher scores on regional attractiveness and international obligations, and is similar in religion and language to the Maldives. We turned our attention to the regional attractiveness score in Figure 14. We found that Singapore, Australia, and Thailand scored lower due to distance, while Ethiopia scored lower due to the local economic level. In terms of international obligations, we believe that those countries that have historically had large

carbon dioxide emissions need to take more responsibility, such as India, Saudi Arabia and Australia in our example.

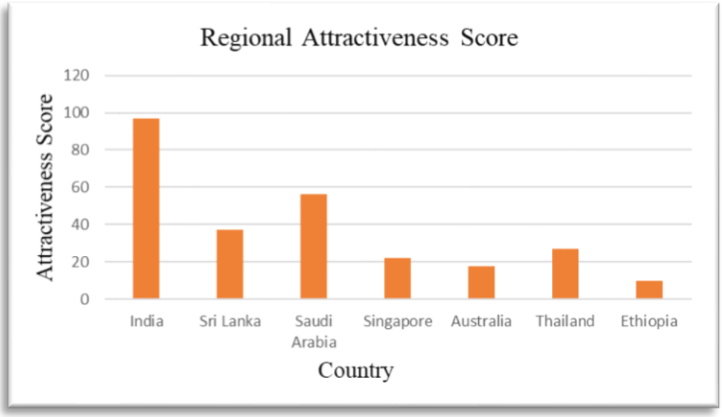


Figure 9. Regional attractiveness score

When we talk about cultural protection, we are mainly talking about religion and language. The majority of the people in the Maldives are Islam-Sunni and speak Dhivehi. But they face environmental immigration. A country with a free religious belief and a diverse language environment is more suitable for them.

5.2 Propose Policies to Address EDPs

Therefore, in the face of severe global EDPs problem, we need to develop a rigorous guidelines for the United Nations to help most island countries that cannot transfer their populations through independent negotiation. Due to the long-term and unexpected nature disasters, our guidelines are also divided into long-term and short-term.

5.2.1 Long-term Guidelines

We divided the long-term guidelines into four major steps, namely the early warning application mechanism, host government screening, the final destination of EDPs and continuous protection.

➢ **Step1: Early Warning Application Mechanism**

- Early warning: The United Nations Marine Inspection Agency can set up relevant departments within the island nation to monitor the sea level and can make accurate forecasts for a period of time in the future. At the same time, the department delineates sea level warning lines for island nations based on forecasts and issues population migration recommendations to island nations 5-10 years in advance.
- Application: The island nations can send an application to the United Nations to transfer the population in advance if they found they are in dangerous.

➢ **Step2: Host government Screening**

- Determine the basic situation and other needs of the countries in need of migration. The United Nations needs to fully understand the number of migrants in the countries in need of migration, the willingness to retain national sovereignty, and whether there are ideal host governments.
- The initial determination of host governments, according to the AOD comprehensive evaluation model established for the United Nations above, a list of host governments can be obtained.
- Further identification of host governments, screening in initially identified countries. First, exclude countries with special and irreconcilable relations: such as political conflicts and economic disputes; second, negotiate and negotiate with the host government on behalf of the

migrating country and the host government, and apply certain international pressure when necessary; Other peaceful countries not on the list can also voluntarily become recipients.

After the above three steps, no more than 5 optional host governments are finally determined, and the following important aspects need to be fully discussed: the acceptable population of the host government, the timing of the island nation's migration, whether the island nation should retain its autonomy, accept Does the country have specific requirements for migrants.

- **Step3: Final Destination of EDPs**

- The United Nations feedbacks the list of host governments to the island nations that submitted the application for immigration. The island nation can make decisions based on its own situation. The island government does not restrict the overall migration, and the island people can choose the countries on the list according to their needs (when the host government agrees). The United Nations can recommend that the island nation try not to disperse its population within a set of conditions. Island nations can also conduct pilot immigration.

- **Step4: Continuous Protection**

- The United Nations should negotiate with the migrating and host governments to sign the corresponding laws and systems in accordance with specific circumstances, while protecting the culture while protecting the human rights of refugees in their new homes. When necessary, the United Nations can also raise funds from countries for relocation and subsequent resettlement in accordance with international obligations.

5.2.2 Short-term Guidelines

Short-term guidelines are specifically designed for unexpected natural disasters, and are applicable when island countries do not have sufficient time to prepare.

In the event of an unexpected natural disaster such as a flood, the island nation can ask the international community for help, and the population can be **temporarily placed** nearby and other countries with the ability to accept the disaster. Because the results of unexpected natural disasters are consistent with long-term natural disasters, they will cause a large number of people to lose their homes. Therefore, the second half of the short-term guidelines are consistent with the long-term guidelines above. Then, we'll present the guidelines more clearly in the form of a diagram:

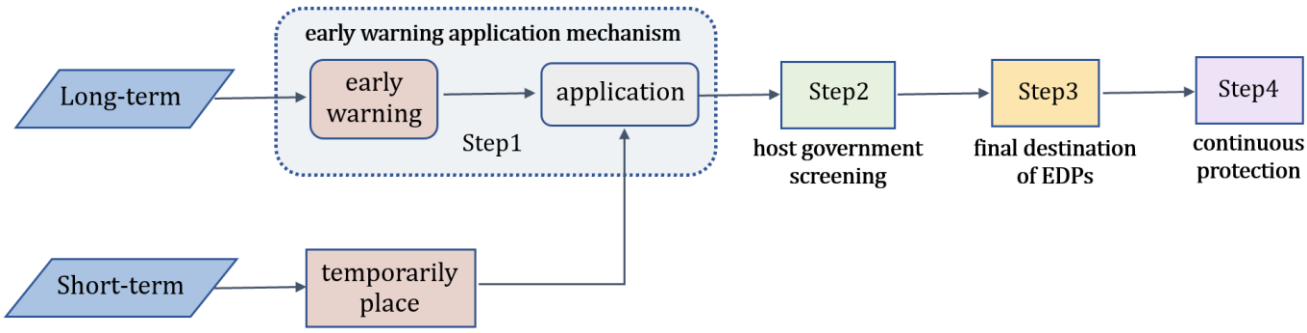


Figure 10. Guidelines for the United Nations

6 Part Three: Potential Impact in Culture

The impact of human rights and cultural differences has been fully considered in our AOD comprehensive evaluation model. However, despite the relatively small cultural differences between the relocated countries and host governments, cultural frictions and conflicts between refugees and

indigenous nationalism are inevitable. These changes will threaten refugees' human rights and their unique culture in a subtle way.

In order to show the phenomenon of cultural communication better, conflict and assimilation, we refined and improved the ideas in the SIR model (Cultural integration model of infectious diseases) [14] to establish a Cultural integration model, which is suitable for cultural communication.

6.1 Cultural Integration Model

The main idea of the Cultural integration model is simulation. First, we set the number of refugees as N_1 and the number of migrants as N_2, and then use a flat square with side a to represent the migrating place, where a is determined by the land area of the migrating place. We use dots to represent people of both cultures and randomly spread them across flat squares. In addition, the following important quantities need to be specified.

- **Status vector**

Let the status vector of person i at time t be $Status_i^t = (x_i^t, y_i^t, c_i^t, class_i)$, x_i^t, y_i^t are position variables, $c_i^t$is the cultural state variable, which represents the change of one's own culture in cultural exchange, and its value fluctuates between 0 and 1, $class_i$ is the initial cultural classification, which is represented by the values 0 and 1 respectively, and does not change with time Variety.

- **Velocity vector**

Let v_i^t represent the direction and magnitude of the speed of person i at time t. During the simulation, the magnitude of the velocity is always fixed, and the direction changes with equal probability in all directions. In particular, the case of removing the boundary is solved by elastic reflection.

$$(x_i^{t+1}, y_i^{t+1}) = (x_i^t, y_i^t) + v_i^t \tag{15}$$

- **Conditions for cultural exchange**

In the simulation process, we need to determine the necessary conditions for cultural communication between people, as follows:

$$\left(x_i^t - x_j^t\right)^2 + \left(y_i^t - y_j^t\right)^2 < {r_w}^2 \tag{16}$$

Where r_w represents the maximum distance of cultural communication, in order to simplify our model we specify that when multiple objects are within distance, communication only occurs between the closest objects.

- **Cultural assimilation**

When person i and person j meet the conditions for cultural exchange, the probability of cultural assimilation P_i for person i is determined solely by $class_i$, which determines whether assimilation occurs. When person i is assimilated, its cultural status will change, which reflects the extent to which assimilation has occurred.

$$P_i = \begin{cases} P_0, & class_i = 0 \\ P_1, & class_i = 1 \end{cases}, c_i^{t+1} = c_i^t + \left(c_j^t - c_i^t\right) \cdot CP \tag{17}$$

Among them, P_0 and P_1 represent the probability of various assimilation of people in the two cultures. It can be seen that when cultural exchange occurs between person i and person j, assimilation does not necessarily occur. CP is the indicator of cultural differences in the AOD model. The greater the cultural differences between individual expressions, the greater the current cultural status gap, and the greater the degree of assimilation.

Using the above model and rules, we can simulate the subtle changes in the culture of the two countries after the migration.

6.2 Simulation

We take the migration from Maldives to India's Hubli as an example for simulation. The population of Maldives is 520,000 and the population of Hubli is 840,000 [15]. We set the corresponding simulation parameters and perform 50 simulations. The following shows the process of cultural change during a simulation.

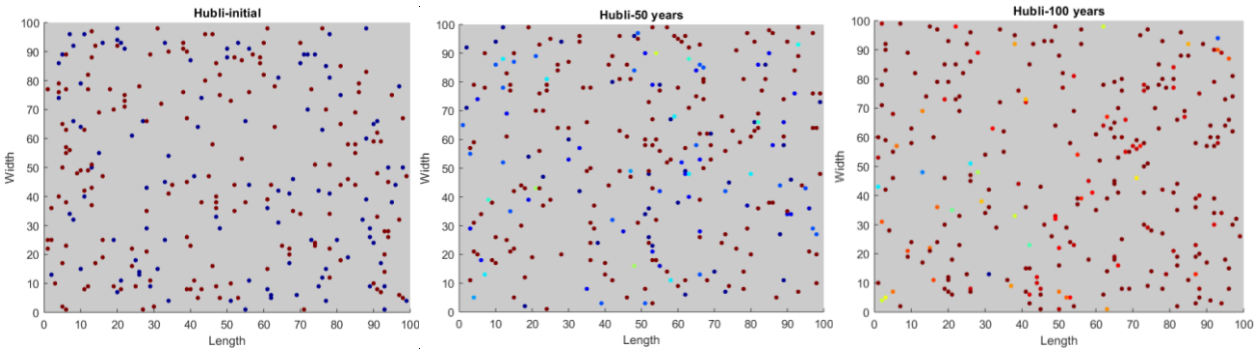


Figure 11. the process of cultural change during a simulation

From the figure, we can find that the culture of Maldives (blue) has gradually been assimilated into Indian culture (red) or blended culture (color) over time. We analyze the culture influences in the process of communication from the following perspectives.

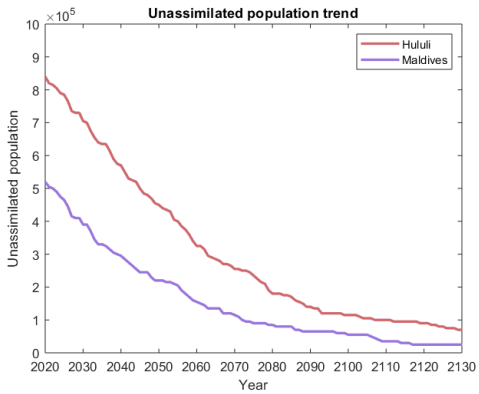


Figure 12. Unassimilated population trend

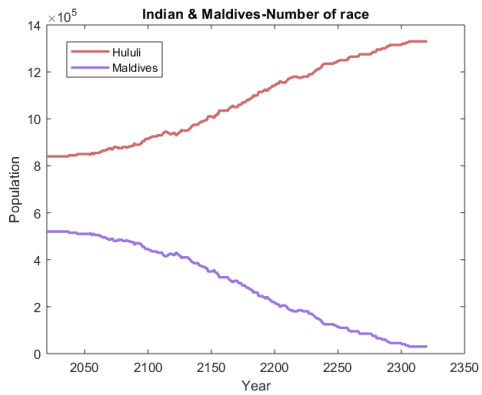


Figure 13. Indian & Maldives-Number of race

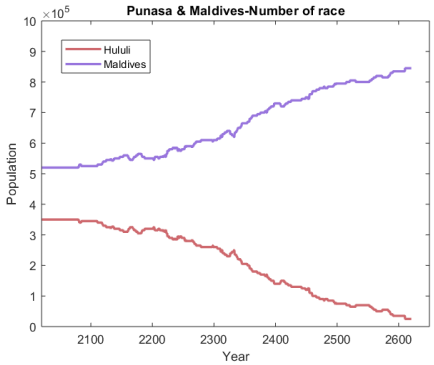
Figure 12 shows that the number of people who have not undergone cultural assimilation (people who are committed to their own culture) is gradually decreasing, and by 2100, more than 85 percent of people will have a blended culture. Figure 13 shows the change of the number of two types of cultural people (with 0.5 as the classification line) over time. It can be clearly seen that the culture of Maldives will be nearly dying around 2300, and after comparing, we can find that it will take more time to completely subvert the culture.

7 Part Four: Policies and Recommendations

In order to protect the culture and human rights of refugees under the sweep of indigenous democracy and culture, some policy guarantees and measures are also necessary after the host government is determined. We use the Cultural integration model to analyze the game between refugee culture and indigenous culture in various situations, and get three effective suggestions.

I. Under the same conditions, refugees should move as far as possible to cities with a population similar to or less than their population.

We migrated the Maldives to two Indian cities Punasa [15] and Nagpur with populations of 35 (10,000) and 222.5 (10,000) respectively, and analyzed them.



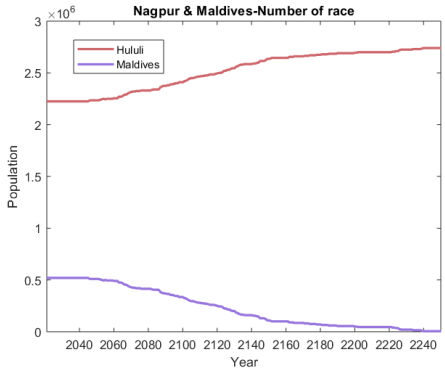


Figure 14. Punasa & Maldives-Number of race **Figure 15. Nagpur & Maldives-Number of race**

The first figure reflects that when refugees migrate to cities with smaller populations, it can promote its own culture while protecting culture. The second figure shows that the Maldivian culture has almost disappeared by about 2200 and is even faster than 2300, which we get in part 3. It shows that when the number of indigenous people is much higher than the migrant population, the demise of culture will come sooner.

II. Group living can protect refugee culture better than scattered living.

We considered the concentration of refugees on one side of the city and still chose Hubli for analysis.

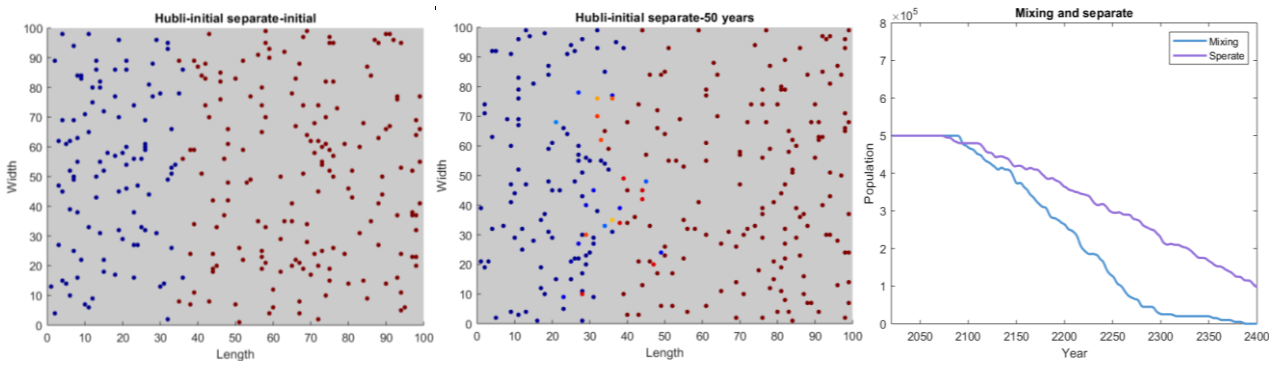


Figure 16. the process of cultural change

The first two pictures reflect the process of cultural integration when living in clusters. The occurrence of assimilation is significantly less, and it mainly occurs at the junction of the place of residence. This is because the opportunity for cultural exchange is greatly reduced by cluster living. From the third picture, it can be found that the period of cultural demise is longer, which has protected the culture of refugees to a certain extent.

III. Improve refugees' national cohesion and cultural self-confidence.

Refugees' cultural self-confidence can be reflected in the probability of cultural assimilation. By adjusting the probability of assimilation, we continue to make analysis of Hubli.

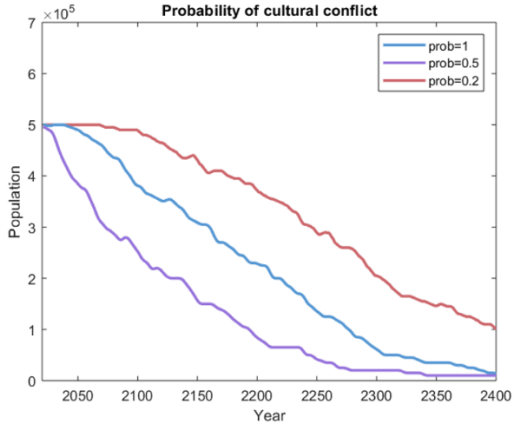


Figure 17. Probability of cultural conflict

It can be seen from the above figure that the smaller the probability of assimilation of refugees in cultural exchanges, the longer their culture will be retained. The probability of assimilation is reflected in the level of national identity and cultural self-confidence. Therefore, it is recommended to take corresponding measures in this regard.

8 Part Five: Importance of Policies

I. Our policy guarantees both individual and collective rights

Individual rights and collective rights are mutually restricted and infiltrated. Only through the realization of collective rights, the necessary economic, cultural and social rights of individuals living in other countries after migration and resettlement of climate refugees can be fully guaranteed. The joint collective life can meet the interests and requirements of individuals if they are determined jointly. Collective rights can better protect individual rights from being violated. Individual rights reflect the realization of collective rights and promote the improvement of collective rights.

On the one hand, this guarantees that each individual climate refugee has the freedom to decide independently whether or not to migrate, how to migrate, the direction and route of migration, and the issue of resettlement. On the other hand, the collective itself is a legal subject that enjoys rights and assumes obligations. The collective interests can be enjoyed by each member of the collective, thereby protecting the rights and interests of individual subjects from the collective level.

II. Our policy combines the capacity for nations to absorb new residents and the obligations due to contributions to climate change

The protection of climate refugees' right to migrate needs to be based on a fair basis in sharing the costs of providing assistance and protection to climate refugees' migration and resettlement on a fair basis. The principle of common but differentiated responsibilities was recognized in the 1992 *Rio Declaration* and fixed in the form of international legal documents. We apply the principle of common but differentiated responsibilities in our AOD model, and in the specific implementation of the protection of the refugees' migration rights, we also quantify economic capacity based on the emissions of the receiving country.

III. Our policy strikes a good balance between protecting the culture of climate-immigrant countries and reducing cultural conflict

The relocated countries facing the risk of disappearing are the countries with their own unique cultures of language, dance, art, music, etc. These unique island cultures are an important part of the cultural diversity of the world. In our policy, we need to give full consideration to the protection of their original culture when arranging to move to a new place of residence in the country. In addition,

we recommend that UNHCR set up a Climate Refugee Assistance Group to serve as an intermediary for coordination and communication between the relocating and receiving countries. We also call on the relevant legislative sections to introduce a Climate Refuge Relocation Law as soon as possible to ensure that every climate refugee can live and work in a new place and reduce the ethnic and ethnic conflicts caused by immigration.

9 Stability and Sensitivity

From the overall effect of our model, it is not only suitable for the environment and region mentioned in this article, but also for most areas related to climate refugees, that is, our model has a relatively broad applicability. After establishing the AOD comprehensive evaluation model of the Climate Refugee Relocation Guide, we have drawn the relationship between the economic level, happiness index, language, culture, and religious beliefs of the host country and the choice of the relocated country. We use the Maldives as an example to study the reliability of this selection guide and get reasonable results. Therefore, our model is practical and reliable. In the comprehensive evaluation model, in order to test whether we will ignore the impact of certain events when we balance the effects of various factors, we try to change the economic level, international obligations, or cultural indicators of a host government by ± 5% and it will not affect the country's candidate ranking. When these indicators change by ± 10%, there will be 1∼2 ranking changes. This also shows that our model has good practicality and reliability.

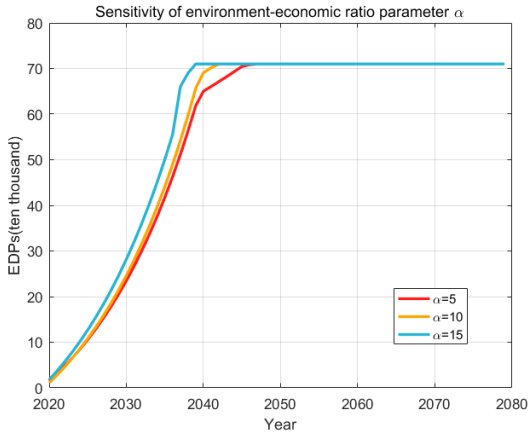


Figure 18. Stability and sensitivity

Change the value of the environment-economic ratio parameter α in the Island Elevation-Resource Model from 5 to 15, that is, change the economic level of a certain country. We find that this has a small effect on the numerical prediction of people facing EDP danger in a certain year, which indicates that our model has very weak sensitivity and it has strong adaptability to countries with different economic levels.

10 Strengths and Weaknesses

10.1 Strengths

- We predict sea level changes at optimistic, retain, and pessimistic levels, which are more practical.
- The AOD comprehensive evaluation system we have established fully reflects the factors of international responsibility, cultural differences, and mutual attraction between migrating and receiving countries, and has high practical significance.

- The TOPSIS ideal point method used in the evaluation of the receiving country makes full use of the index data and avoids the disadvantages of subjective weights.
- Analyze the blending process of culture in a simulated way, so that the demise of culture can be quantified and intuitively reflected.

10.2 Weaknesses

- We use the data of sea level changes to make predictions instead of dynamic models including climate change. Although some accuracy is lost, the model is simplified.
- Cultural differences may be difficult to express comprehensively in the process of quantification, but such distortions as culturally complex things are unavoidable.

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Appendix

1、Calculation of the average ocean level change

```
ncFilePath='dt_global_allsat_msla_h_y2019_m04.nc';lon=ncread(ncFilePath,'lon');%Read longitude variable
lat=ncread(ncFilePath,'lat');%Read latitude variable
time=ncread(ncFilePath,'time');%Read time variable
sla=ncread(ncFilePath,'sla');%Av
y(1,1)=nansum(nansum(sla(1:500,:)))/(numel(sla(1:500,:))-numel(find(isnan(sla(1:500,:)))));
y(1,2)=nansum(nansum(sla(501:1000,:)))/(numel(sla(501:1000,:))-numel(find(isnan(sla(501:1000,:)))));
y(1,3)=nansum(nansum(sla(1001:1440,:)))/(numel(sla(1001:1440,:))-numel(find(isnan(sla(1001:1440,:)))));
%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

2、

```
%Setting the axes and background
axes1 = axes('xAxisLocation','top','yAxisLocation','right');
hold(axes1,'all');
axes2 = axes('xAxisLocation','bottom','yAxisLocation','left');
box(axes2,'on')
set(axes2,'xtick',2000:10:2060)
set(axes2,'xlim',[2005,2060])
set(axes2,'ytick',[-2:2:18])
set(axes2,'ylim',[-2,18])
set(axes1,'xtick',[])
set(axes1,'xlim',[2005,2060])
set(axes1,'ytick',[])
set(axes1,'ylim',[-2,18])
set(axes2,'LineWidth',1.5)
set(axes1,'LineWidth',1.5)
hold on
whitebg([243/256,244/256,246/256])
hold on

%Raw data
hpm=hpm*10;
```

```
year1=2005:1:2019;
hpm2=hpm+0.3;
hpm3=hpm-0.3;
hpm2=hpm2';
hpm3=hpm3';
for i=1:length(year1)
    year2(i)=year1(length(year1)-i+1);
    hpm4(i)=hpm3(length(year1)-i+1);
end
hold on
plot(year1,hpm,'k','LineWidth',3,'MarkerSize',3)

%Fitting for optimistic situations
yearp1=2019:2060;
hpmp=ones(1,42)*hpm(15);
hpmp2=hpmp+0.3;
hpmp3=hpmp-0.3;
for i=1:length(yearp1)
    yearp2(i)=yearp1(length(yearp1)-i+1);
    hpmp4(i)=hpmp3(length(yearp1)-i+1);
end
hold on
plot(yearp1,hpmp,'color',[152/256,117/256,221/256],'LineWidth',3,'MarkerSize',3)

%Linear fit for intermediate cases
yearx=1:length(hpm);
yearx=yearx';
hpmx=hpm;
X=[ones(length(hpmx),1),yearx];
[b2,bint2,r2,rint2,stats2]=regress(hpmx,X,0.1);
yearx1=2019:2060;
hpmx1=b2(2)*(yearx1-2004)+b2(1);
hpmx2=bint2(2,1)*(yearx1-2004)+bint2(1,1);
hpmx3=bint2(2,2)*(yearx1-2004)+bint2(1,2);
for i=1:length(yearx1)
```

```
    yearx2(i)=yearx1(length(yearx1)-i+1);
    hpmx4(i)=hpmx3(length(yearx1)-i+1);
end
hold on
plot(yearx1,hpmx1,'color',[87/256,153/256,213/256],'LineWidth',3,'MarkerSize',3)
%Exponential fitting for pessimistic situations
yearz=1:length(hpm);
yearz=yearz';
hpmz=log(hpm);
X=[ones(length(hpmz),1),yearz];
[b,bint,r,rint,stats]=regress(hpmz,X,0.5);
pc=sum(abs(r))./length(r);
a=exp(b(1));
yearz1=2019:2060;
hpmz1=a.*exp(b(2).*(yearz1-2004));
a1=exp(bint(1,1));
a2=exp(bint(1,2));
hpmz2=a1.*exp(bint(2,1).*(yearz1-2004));
hpmz3=a2.*exp(bint(2,2).*(yearz1-2004));
for i=1:length(yearz1)
    yearz2(i)=yearz1(length(yearz1)-i+1);
    hpmz4(i)=hpmz3(length(yearz1)-i+1);
end
hold on
plot(yearz1,hpmz1,'color',[206/256,104/256,109/256],'LineWidth',3,'MarkerSize',3)

%Graphic drawing
hold on
fill([year1,year2],[hpm2,hpm4],[203/256,203/256,203/256],'EdgeColor',[203/256,203/256,203/256])
hold on
plot(year1,hpm,'k','LineWidth',3,'MarkerSize',3)
hold on
fill([yearp1,yearp2],[hpmp2,hpmp4],[192/256,178/256,236/256],'EdgeColor',[192/256,178/256,236/
256])
hold on
```

```
plot(yearp1,hpmp,'color',[152/256,117/256,221/256],'LineWidth',3,'MarkerSize',3)
hold on
fill([yearx1,yearx2],[hpmx2,hpmx4],[172/256,204/256,234/256],'EdgeColor',[172/256,204/256,234/
256])
hold on
plot(yearx1,hpmx1,'color',[87/256,153/256,213/256],'LineWidth',3,'MarkerSize',3)
hold on
fill([yearz1,yearz2],[hpmz2,hpmz4],[223/256,162/256,161/256],'EdgeColor',[223/256,162/256,161/2
56])
hold on
plot(yearz1,hpmz1,'color',[206/256,104/256,109/256],'LineWidth',3,'MarkerSize',3)
hold on
xf=[2019,2019];
yf=[-2,18];
hold on
plot(xf,yf,'k','LineWidth',1)
legend('Raw data','Optimistic','Retain','Pessimistic')
xlabel('Year')
ylabel('Global sea level anomalies(cm)')
%%%
```

```
end
w=[yy,zj,gdp];
b=[1,1,1];
phi=3.14*30/180;
for i=1:7
    mc(i)=norm(w(i,:),2);
    jc(i)=acos(dot(w(i,:),b)/(norm(w(i,:))*norm(b)));
    cp(i)=mc(i)*(1-cos(jc(i)+phi));
end
maxxyl=max(xyl);
minxyl=min(xyl);
maxco2=max(co2);
minco2=min(co2);
maxcp=max(cp);
mincp=min(cp);
for i=1:7
    xyl(i)=(xyl(i)-minxyl)/(maxxyl-minxyl);
    co2(i)=(co2(i)-minco2)/(maxco2-minco2);
    cp(i)=(maxcp-cp(i))/(maxcp-mincp);
end
s=1;
for i=1:7
    xyl(i)=xyl(i)/sqrt(sum(xyl.^2));
    co2(i)=co2(i)/sqrt(sum(co2.^2));
    cp(i)=s*cp(i)/sqrt(sum(cp.^2));
end
vmax1=max(xyl);
vmax2=max(co2);
vmax3=max(cp);
vmin1=min(xyl);
vmin2=min(co2);
vmin3=min(cp);
for i=1:7
   d1(i)=sqrt((xyl(i)-vmax1)^2+(co2(i)-vmax2)^2+(cp(i)-vmax3)^2);
   d2(i)=sqrt((xyl(i)-vmin1)^2+(co2(i)-vmin2)^2+(cp(i)-vmin3)^2);
```

```
  socre(i)=d2(i)/(d1(i)+d2(i));
end

%%%%%%%%%%%%%%%%%%%%%%%%%%
4、%plot islands nations position
ax = worldmap('World');
setm(ax, 'Origin', [0 180 0])
land = shaperead('landareas', 'UseGeoCoords', true);
geoshow(ax, land, 'FaceColor', [137/255 157/255 192/255]);

%Maldives coordinates
MalLon = 73.5;%longitude;
MalLat = 4.2;%latitude
%use red. to Callout
geoshow(MalLat, MalLon, 'Marker','.','markersize',10,'MarkerEdgeColor','red')
textm(MalLat,MalLon +1, 'Maldives');
%Marshall Islands coordinates
MarLon = 171.1;%longitude;
MarLat = 7.1;%latitude
%use red. to Callout
geoshow(MarLat, MarLon, 'Marker','.','markersize',10,'MarkerEdgeColor','red')
textm(MarLat,MarLon +1, 'Marshall Islands');
%Tuvalu coordinates
TuvLon = 179.1;%longitude;
TuvLat = -8.3;%latitude
%use red. to Callout
geoshow(TuvLat, TuvLon, 'Marker','.','markersize',10,'MarkerEdgeColor','red')
textm(TuvLat+2.5,TuvLon , 'Tuvalu');
%Kiribati coordinates
KirLon = 172.5;%longitude;
KirLat = 1.25;%latitude
%use red* to Callout
geoshow(KirLat, KirLon, 'Marker','.','markersize',10,'MarkerEdgeColor','red')
textm(KirLat,KirLon + 1, 'Kiribati');
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%
5、calculate the dangerous sea level
h=1.2;g=1;
h_dangerous=0.75*0.75/((1+1/h)*(1-g/10));
indian_add=zeros(3,62);
indian_add(1,1)=0;indian_add(2,1)=0;indian_add(3,1)=0;
pacific_add=zeros(3,62);
pacific_add(1,1)=0;pacific_add(2,1)=0;pacific_add(3,1)=0;
atlantic_add=zeros(3,62);
atlantic_add(1,1)=0;atlantic_add(2,1)=0;atlantic_add(3,1)=0;
for i=1:3
   for j=1:61
      indian_add(i,j+1)=indian_add(i,j)+indian(i,j+1);
      pacific_add(i,j+1)=pacific_add(i,j)+pacific(i,j+1);
      atlantic_add(i,j+1)=atlantic_add(i,j)+atlantic(i,j+1);
   end
end
x=2020:2080;
x=x';
figure(1)
plot(x,indian_add(1,2:end),'Color',[152/255 117/255 221/255],'Linewidth',2);hold on;
plot(x,indian_add(2,2:end),'Color',[87/255 153/255 213/255],'Linewidth',2);hold on;
plot(x,indian_add(3,2:end),'Color',[206/255 104/255 109/255],'Linewidth',2);
legend('sla unchanged, sl rises linearly','sla increases linearly, sl accelerates linearly','sla index
increases, sl explodes');hold on;
title('indian sea level predict');
figure(2)
plot(x,pacific_add(1,2:end),'Color',[152/255 117/255 221/255],'Linewidth',2);hold on;
plot(x,pacific_add(2,2:end),'Color',[87/255 153/255 213/255],'Linewidth',2);hold on;
plot(x,pacific_add(3,2:end),'Color',[206/255 104/255 109/255],'Linewidth',2);
legend('sla unchanged, sl rises linearly','sla increases linearly, sl accelerates linearly','sla index
increases, sl explodes');hold on;
title('pacific sea level predict');
figure(3)
plot(x,atlantic_add(1,2:end),'Color',[152/255 117/255 221/255],'Linewidth',2);hold on;
plot(x,atlantic_add(2,2:end),'Color',[87/255 153/255 213/255],'Linewidth',2);hold on;
```

```
plot(x,atlantic_add(3,2:end),'Color',[206/255 104/255 109/255],'Linewidth',2);
legend('sla unchanged, sl rises linearly','sla increases linearly, sl accelerates linearly','sla index increases, sl explodes');hold on;
% title('atlantic sea level predict');

%%%
```

```
    edp_pop33=min(rate_edp33*pop(3),pop(3));
    %
    rate_edp41=pacific_add(1,yeari+1)*(1+1/hh(4))*(1-pg(4)/10)/sll(4);  %pacific_add(1,i) maldives
    edp_pop41=min(rate_edp41*pop(4),pop(4));
    rate_edp42=pacific_add(2,yeari+1)*(1+1/hh(4))*(1-pg(4)/10)/sll(4);  %pacific_add(1,i) maldives
    edp_pop42=min(rate_edp42*pop(4),pop(4));
    rate_edp43=pacific_add(3,yeari+1)*(1+1/hh(4))*(1-pg(4)/10)/sll(4);  %pacific_add(1,i) maldives
    edp_pop43=min(rate_edp43*pop(4),pop(4));

    %
    edp_total(1,yeari)=edp_pop11+edp_pop21+edp_pop31+edp_pop41;
    edp_total(2,yeari)=edp_pop12+edp_pop22+edp_pop32+edp_pop42;
    edp_total(3,yeari)=edp_pop13+edp_pop23+edp_pop33+edp_pop43;
end
 x=x';
plot(x(1,1:60),edp_total(1,1:end),'Color',[152/255 117/255 221/255],'Linewidth',2);hold on;
plot(x(1,1:60),edp_total(2,1:end),'Color',[87/255 153/255 213/255],'Linewidth',2);hold on;
plot(x(1,1:60),edp_total(3,1:end),'Color',[206/255 104/255 109/255],'Linewidth',2);
legend('Optimistic','Retain','Pessimistic');
xlabel('Year');
ylabel('EDP population(ten thousand)');
title('EDP population');
axis([2020 2085,0,120])

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
7、Cultural conflict simulation
pe=100;pn=168;
x=zeros(pe+pn,1);
y=zeros(pe+pn,1);
x(1:pe,1)= randi([1 100],pe,1);
y(1:pe,1)= randi([1 100],pe,1);
x((pe+1):(pe+pn),1)= randi([100 100],pn,1);
y((pe+1):(pe+pn),1)= randi([1 100],pn,1);
z=zeros(pe+pn,1);
z((pe+1):(pe+pn),1)=1;
```

```
alpha=0.1;
ppp=3000;
for bi=1:ppp
    for i=1:(pe+pn)
    x(i,1)=x(i,1)+randi([-1 1]);
    y(i,1)=y(i,1)+randi([-1 1]);

        if x(i,1)<=0
            x(i,1)=1;
        end
        if x(i,1)>=100
            x(i,1)=99;
        end
        if y(i,1)<=0
            y(i,1)=1;
        end
        if y(i,1)>=100
            y(i,1)=99;
        end

    end

    for i=1:(pe+pn)
        xm=x(i,1);
        ym=y(i,1);
        zm=z(i,1);
        for j=(i+1):(pe+pn)
            if xm==x(j,1) && ym==y(j,1) && zm~=z(j,1)
                zn=z(j,1);
                z(i,1)=zm+(zn-zm)*alpha;
                z(j,1)=zn+(zm-zn)*alpha;
            end
        end
    end
    if mod(bi,4)==0
```

```
    az=find(z<=0.5);   %1pe 2pn
    true_peo(1,bi/4)=length(az);
    az=find(z>0.5);   %1pe 2pn
    true_peo(2,bi/4)=length(az);
    end
end
S = 15;
scatter(x,y,S,z,'filled')
set(gca, 'color', [203/255 203/255 203/255])
xlabel('Length')
ylabel('Width')
title('Hubli-initial separate-initial')
grid off
colormap(jet)
xlim([0 100])
ylim([0 100])
```